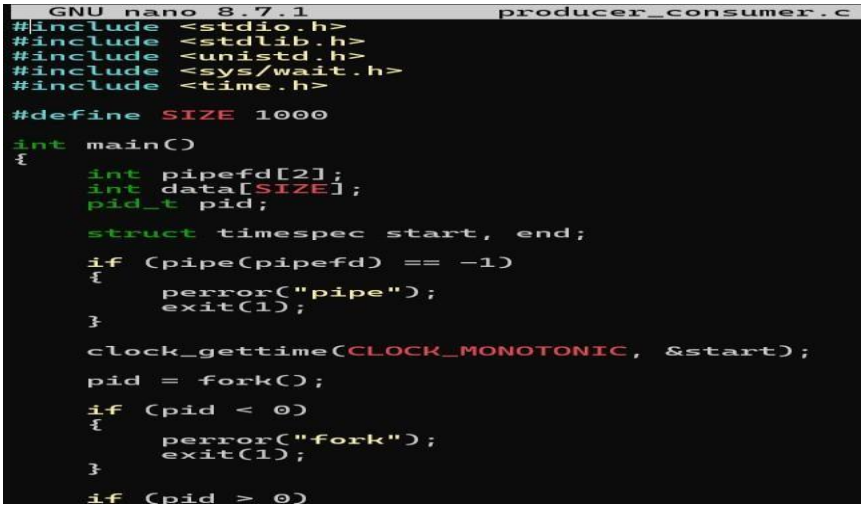
PRACTICAL WEEK 5

Task:-1

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Source code:-



```
  GNU nano 8.7.1                producer_consumer.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

#define SIZE 1000

int main()
{
    int pipefd[2];
    int data[SIZE];
    pid_t pid;

    struct timespec start, end;

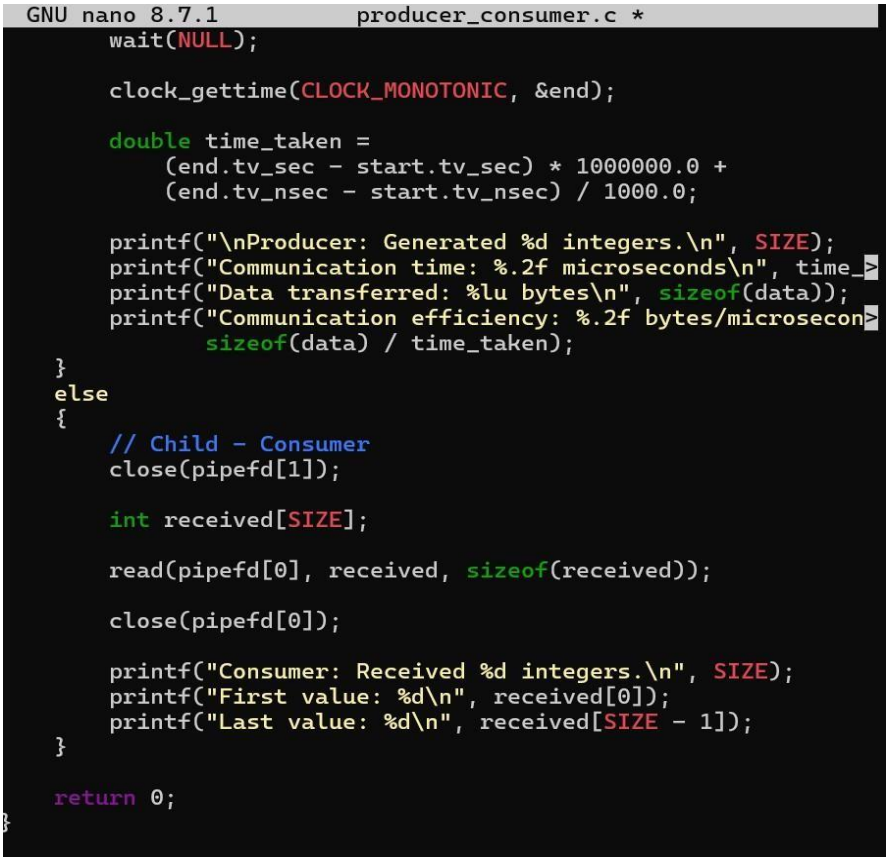
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    clock_gettime(CLOCK_MONOTONIC, &start);

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        exit(1);
    }

    if (pid > 0)
```



```
GNU nano 8.7.1          producer_consumer.c *
        wait(NULL);

        clock_gettime(CLOCK_MONOTONIC, &end);

        double time_taken =
            (end.tv_sec - start.tv_sec) * 1000000.0 +
            (end.tv_nsec - start.tv_nsec) / 1000.0;

        printf("\nProducer: Generated %d integers.\n", SIZE);
        printf("Communication time: %.2f microseconds\n", time_>
        printf("Data transferred: %lu bytes\n", sizeof(data));
        printf("Communication efficiency: %.2f bytes/microsecon>
               sizeof(data) / time_taken);
    }
    else
    {
        // Child - Consumer
        close(pipefd[1]);

        int received[SIZE];

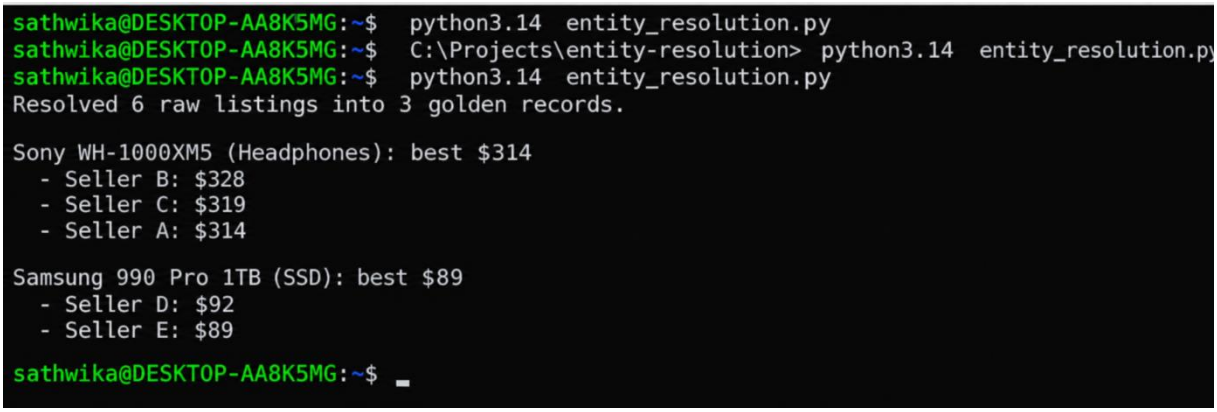
        read(pipefd[0], received, sizeof(received));

        close(pipefd[0]);

        printf("Consumer: Received %d integers.\n", SIZE);
        printf("First value: %d\n", received[0]);
        printf("Last value: %d\n", received[SIZE - 1]);
    }

    return 0;
}
```

OUTPUT:-


sathwika@DESKTOP-AA8K5MG:~$ python3.14 entity_resolution.py
sathwika@DESKTOP-AA8K5MG:~$ C:\Projects\entity-resolution> python3.14 entity_resolution.py
sathwika@DESKTOP-AA8K5MG:~$ python3.14 entity_resolution.py
Resolved 6 raw listings into 3 golden records.
Sony WH-1000XM5 (Headphones): best $314
- Seller B: $328
- Seller C: $319
- Seller A: $314
Samsung 990 Pro 1TB (SSD): best $89
- Seller D: $92
- Seller E: $89
sathwika@DESKTOP-AA8K5MG:~$

The parent process acts as the **producer** and sends data through an anonymous pipe. The child process acts as the **consumer** and reads the data from the pipe. The communication time and data transfer rate are measured to evaluate the efficiency of inter-process communication.

Observation

Task:-2

Develop a program that executes the equivalent of the shell command:

ls -l | grep ".c" using fork(), pipe(), dup2(), and

exec() system calls.

Source code:-

```
  GNU nano 8.7.1                  pipe_exec.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    // First child: ls -l
    pid1 = fork();

    if (pid1 < 0)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        // Redirect stdout to pipe
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);
```

```
 GNU nano 8.7.1                pipe_exec.c *

    if (pid2 < 0)
    {
        perror("fork");
        exit(1);
    }

    if (pid2 == 0)
    {
        // Redirect stdin from pipe
        dup2(pipefd[0], STDIN_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        execlp("grep", "grep", ".c", NULL);

        perror("execlp grep");
        exit(1);
    }

    // Parent closes pipe
    close(pipefd[0]);
    close(pipefd[1]);

    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);

    printf("\nCommand executed: ls -l | grep \".c\"\n");

    return 0;
}
```

OUTPUT

```
sathwika@DESKTOP-AA8K5MG:~$ nano pipe_exec.c
sathwika@DESKTOP-AA8K5MG:~$ gcc pipe_exec.c -o pipe_exec
sathwika@DESKTOP-AA8K5MG:~$ ./pipe_exec
-rw-r--r--  1 sathwika sathwika    16628 Aug 11 06:59 command_executor
-rw-r--r--  1 sathwika sathwika     1387 Aug 11 06:59 command_executor.c
-rw-r--r--  1 sathwika sathwika    10000 Aug 11 06:41 doc
-rw-r--r--  1 sathwika sathwika    16256 Aug  7 05:10 file_copy
-rw-r--r--  1 sathwika sathwika     5113 Aug  7 05:17 file_copy.c
-rw-r--r--  1 sathwika sathwika    16432 Aug 13 08:28 fork_exec
-rwxr-xr-x  1 sathwika sathwika     4020 Aug  7 06:17 fork_exec.c
-rw-r--r--  1 sathwika sathwika     4006 Aug  7 05:12 include
-rw-r--r--  1 sathwika sathwika     5098 Aug  7 05:19 keyboard_input.c
-rw-r--r--  1 sathwika sathwika      800 Aug  7 05:09 main_loop.c
-rwxr-xr-x  1 sathwika sathwika     3924 Aug  7 06:11 parser_double_quote.c
sathwika@DESKTOP-AA8K5MG:~$ ls -l
-rwxr-xr-x  1 sathwika sathwika     2709 Aug 11 06:38 parser_quotes.c
-rw-r--r--  1 sathwika sathwika    16344 Sep 10 08:33 pipe_exec
-rw-r--r--  1 sathwika sathwika     1189 Sep 10 08:33 pipe_exec.c
-rw-r--r--  1 sathwika sathwika    15952 Aug  1 04:09 process
-rw-r--r--  1 sathwika sathwika     2791 Aug  1 04:08 process.c
-rw-r--r--  1 sathwika sathwika     8741 Aug 11 10:15 producer_consumer
-rw-r--r--  1 sathwika sathwika    16408 Sep 10 08:31 producer_consumer.c
-rw-r--r--  1 sathwika sathwika     1666 Sep 10 08:30 producer_consumer.c~
-rw-r--r--  2 sathwika sathwika     4096 Aug  5 09:14 script2
-rw-r--r--  1 sathwika sathwika      559 Aug 13 12:00 session2_task3.c
-rw-r--r--  1 sathwika sathwika      837 Aug 13 12:10 session2_task4.c
-rw-r--r--  1 sathwika sathwika     4635 Aug 13 12:22 session2_task2.c
-rw-r--r--  1 sathwika sathwika     3160 Aug 15 12:34 session2_task1.c
-rw-r--r--  1 sathwika sathwika     1279 Aug 15 12:34 session1_task2.c
-rw-r--r--  1 sathwika sathwika     2721 Aug 15 12:42 session1_task1.c
-rw-r--r--  1 sathwika sathwika     2608 Aug 16 12:32 session0_task2.c
-rw-r--r--  1 sathwika sathwika     1217 Aug 15 13:57 session0_task1.c
-rw-r--r--  1 sathwika sathwika     1005 Aug 15 13:59 session0.c
-rw-r--r--  1 sathwika sathwika     2283 Aug 15 13:07 session7_task1.c
-rw-r--r--  1 sathwika sathwika     1448 Aug 15 13:27 session7_task2.c
-rw-r--r--  1 sathwika sathwika     3280 Aug 15 13:15 session8_task1.c
-rw-r--r--  1 sathwika sathwika     3602 Aug 15 13:12 session8_task2.c
-rw-r--r--  1 sathwika sathwika      879 Aug 13 10:13 skill_week1_task2
sathwika@DESKTOP-AA8K5MG:~$
```

Observation

The program successfully implements the shell pipeline ls -l | grep ".c" using Linux system calls. The first child executes ls -l and sends its output through the pipe. The second child receives the output through the pipe and executes grep ".c" to display C-related files